

HR EMPLOYEE ATTRITION ANALYSIS

DATASET

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

TOOLS USED

- SQL
- Power BI
- MS Word

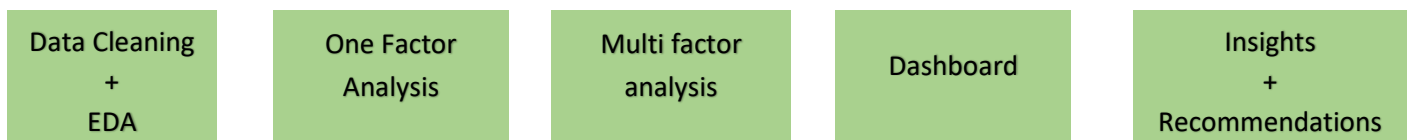
PROBLEM STATEMENT

Employee attrition leads to increased recruitment costs, loss of experienced talent, and reduced organizational productivity. The goal is to identify the key factors influencing employee attrition and determine the employee segments at the highest risk of leaving, enabling the development of targeted retention strategies.

KEY OBJECTIVES

1. Measure the overall employee attrition rate.
2. Identify departments and job roles with the highest turnover.
3. Assess the impact of salary, overtime, business, travel and other factors on attrition.
4. Identify high-risk employee segments by combining multiple factors.
5. Provide actionable recommendations to improve employee retention.

WORKFLOW



STEPS INVOLVED IN DATA PREPARATION

1. Imported raw data into MySQL.
2. Created derived columns
 - Age Band
 - Salary Slab
 - Distance Band
 - Years at Company Band
3. Built a reusable SQL view (employee_profile) for reporting.
4. Connected the SQL view to Power BI for interactive visualization and reporting.

1. Overall Attrition Rate

```
50 -- Employee status
51 • SELECT COUNT(EmployeeNumber) AS EmpCount,
52    CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1),'%') AS Attrition_rate
53 FROM employee_profile;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	EmpCount	Attrition_rate		
▶	1470	16.1%		

2. Department-wise Attrition

```
54 -- Departmen wise Attrition
55 • SELECT Department,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
56    CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1),'%') AS Attrition_rate
57 FROM employee_profile
58 GROUP BY Department;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	Department	Attr_cnt	tot_emp	Attrition_rate
	Sales	92	446	20.6%
	Research & Development	133	961	13.8%
	Human Resources	12	63	19.0%

- Sales and HR departments have higher attrition rates than R&D.

Takeaway: Sales in particular looks like a “hot spot” for turnover and may require targeted interventions (compensation, quotas, management style).

3. Attrition by Job Role

```
59 -- Attrition by Job Role
60 • SELECT JobRole,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
61    CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1),'%') AS Attrition_rate
62 FROM employee_profile
63 GROUP BY JobRole
64 ORDER BY Attrition_rate;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
	JobRole	Attr_cnt	tot_emp	Attrition_rate
▶	Research Scientist	47	292	16.1%
	Sales Executive	57	326	17.5%
	Research Director	2	80	2.5%
	Human Resources	12	52	23.1%
	Laboratory Technician	62	259	23.9%
	Sales Representative	33	83	39.8%
	Manager	5	102	4.9%
	Manufacturing Director	10	145	6.9%
	Healthcare Representative	9	131	6.9%

- Sales Representative is the single highest-risk role (~40% attrition).
- Lab Technicians and HR staff are also significantly above the company average.
- Senior roles (Managers, Directors) are relatively stable.

Takeaway: These roles should be top priority for retention efforts.

4. Attrition by Gender

```
64 -- Attrition by Gender
65 • SELECT Gender, SUM(Attrition='Yes') as Attr_cnt, COUNT(EmployeeNumber) as tot_emp,
66     CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1), '%') AS Attrition_rate
67 FROM employee_profile
68 GROUP BY Gender;
```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	Gender	Attr_cnt	tot_emp	Attrition_rate
▶	Female	87	588	14.8%
	Male	150	882	17.0%

- Male employees have a slightly higher attrition rate than females
- Gender alone is not a strong determinant as the difference is relatively small.

5. Attrition by Age

```
69 -- Age Group wise Attrition
70 • SELECT Age_band, SUM(Attrition = 'Yes') AS Attr_cnt, COUNT(EmployeeNumber) AS total_cnt,
71     CONCAT(ROUND(SUM(Attrition = 'Yes')/COUNT(EmployeeNumber)*100,1), '%') AS Attr_Rate
72 FROM employee_profile
73 GROUP BY Age_band;
```

Result Grid				
		Filter Rows:	Export:	Wrap Cell Content:
	Age_Band	Attr_cnt	total_cnt	Attr_Rate
▶	35-44	51	505	10.1%
	45-54	25	245	10.2%
	25-34	112	554	20.2%
	55+	11	69	15.9%
	<25	38	97	39.2%

- Attrition is highest among employees under 25.
- Attrition remains elevated in the 25–34 age group, then drops significantly from 35–54.
- There is a moderate uptick again at 55+, but the main risk is clearly early-career staff.

Takeaway: This suggests early tenure and early career expectations are not being met.

6. Attrition by Overtime

```
74 -- Attrition by Overtime
75 • SELECT OverTime,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
76    CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1), '%') AS Attrition_rate
77 FROM employee_profile
78 GROUP BY OverTime;
```

	OverTime	Attr_cnt	tot_emp	Attrition_rate
▶	Yes	127	416	30.5%
	No	110	1054	10.4%

- Employees working overtime have a rate of 30.5%, nearly three times higher than employees who do not work overtime (10.4%).

Takeaway: Excessive workload and longer working hours are strongly associated with higher employee turnover.

7. Attrition by Travel frequency

```
79 -- Attrition by travel frequency
80 • SELECT BusinessTravel,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
81    CONCAT(ROUND(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, 1), '%') AS Attrition_rate
82 FROM employee_profile
83 GROUP BY BusinessTravel;
```

	BusinessTravel	Attr_cnt	tot_emp	Attrition_rate
▶	Travel_Rarely	156	1043	15.0%
	Travel_Frequently	69	277	24.9%
	Non-Travel	12	150	8.0%

- Employees who travel frequently have the highest attrition rate (24.9%), followed by those who travel rarely (15.0%)
- Employees with no business travel have the lowest attrition rate (8.0%)

Takeaway: Optimize business travel by replacing non-essential trips with virtual meetings and support hybrid or WFH arrangements.

8. Attrition by Distance

```
84 -- Attrition by distance
85 • SELECT Distance_band, SUM(Attrition = 'Yes') AS Attr_cnt, COUNT(EmployeeNumber) AS total_cnt,
86    CONCAT(ROUND(SUM(Attrition = 'Yes')/COUNT(EmployeeNumber)*100,1), '%') AS Attr_Rate
87 FROM employee_profile
88 GROUP BY Distance_band
89 ORDER BY Distance_band;
```

Distance_Band	Attr_cnt	total_cnt	Attr_Rate
<10	133	940	14.1%
10-20	59	326	18.1%
20-30	45	204	22.1%

- Attrition increases as distance from home increases.
- Employees living 20–30 km from the workplace have the highest attrition rate.

Takeaway: HR can consider hybrid or WFH options or transportation support to improve employee retention.

9. Attrition by Number of Years at company

```
90 -- Attrition by years at company
91 • SELECT YAC_band, SUM(Attrition = 'Yes') AS Attr_cnt, COUNT(EmployeeNumber) AS total_cnt,
92    CONCAT(ROUND(SUM(Attrition = 'Yes')/COUNT(EmployeeNumber)*100,1), '%') AS Attr_Rate
93 FROM employee_profile
94 GROUP BY YAC_band
95 ORDER BY YAC_band;
```

YAC_band	Attr_cnt	total_cnt	Attr_Rate
<10	199	1104	18.0%
10-20	30	300	10.0%
30+	8	66	12.1%

- Attrition is highest in the first few years (roughly 0–3 years) at the company.
- After ~ 4–5 years, the number of leavers declines sharply.
- Very long-tenured employees (10+ years) have comparatively low attrition.

Takeaway: Most employees who leave do so within their first few years, often before they become fully embedded in the organization. This is consistent with the age band finding.

10. Attrition by work life Balance

```
101 -- Attrition by work life balance
102 • SELECT WorkLifeBalance,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
103      CONCAT(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, '%') AS Attrition_rate
104 FROM employee_profile
105 GROUP BY WorkLifeBalance;
```

	WorkLifeBalance	Attr_cnt	tot_emp	Attrition_rate
▶	1	25	80	31.2500%
	3	127	893	14.2217%
	2	58	344	16.8605%
	4	27	153	17.6471%

- Employees with the poorest work-life balance (Level 1) have the highest attrition rate (31.3%), more than double that of employees with Level 3 (14.2%).
- Employees reporting a better work-life balance tend to stay longer.

Takeaway: Promoting flexible work arrangements, manageable workloads, and employee well-being initiatives can help improve work-life balance

11. Attrition by Relationship satisfaction

```
106 -- Attrition by Relationship satisfaction
107 • SELECT RelationshipSatisfaction,SUM(Attrition='Yes') as Attr_cnt,COUNT(EmployeeNumber) as tot_emp,
108      CONCAT(SUM(Attrition='Yes')/COUNT(EmployeeNumber)*100, '%') AS Attrition_rate
109 FROM employee_profile
110 GROUP BY RelationshipSatisfaction;
```

	RelationshipSatisfaction	Attr_cnt	tot_emp	Attrition_rate
▶	1	57	276	20.6522%
	4	64	432	14.8148%
	2	45	303	14.8515%
	3	71	459	15.4684%

- Employees with the lowest relationship satisfaction (Level 1) have the highest attrition rate (20.7%).
- Attrition remains fairly consistent (14.8%–15.5%) among employees with moderate to high relationship satisfaction.

Takeaway: Employees with poor workplace relationships are more likely to leave, while improvements beyond a moderate level of relationship satisfaction show only marginal differences in attrition.

12. Attrition by Salary

```

111  -- Attrition by Salary
112  •  SELECT Salary_slab, SUM(Attrition = 'Yes') AS Attr_cnt, COUNT(EmployeeNumber) AS total_cnt,
113     CONCAT(ROUND(SUM(Attrition = 'Yes')/COUNT(EmployeeNumber)*100,1), '%') AS Attr_Rate
114  FROM employee_profile
115  GROUP BY Salary_slab
116  ORDER BY
117     CASE
118         WHEN Salary_slab = '<5000' THEN 1
119         WHEN Salary_slab = '5000-9999' THEN 2
120         WHEN Salary_slab = '10000-14999' THEN 3
121         WHEN Salary_slab = '15000+' THEN 4
122     END;

```

Result Grid					Filter Rows:		Export:	Wrap Cell Content:
	Salary_Slab	Attr_cnt	total_cnt	Attr_Rate				
▶	<5000	163	749	21.8%				
	5000-9999	49	440	11.1%				
	10000-14999	20	148	13.5%				
	15000+	5	133	3.8%				

- Employees earning less than 5000 have the highest attrition rate (21.8%), while those earning 15000 or more have the lowest attrition rate (3.8%).

Takeaway: Attrition tends to decrease as monthly income increases, indicating that compensation plays an important role in employee retention.

INSIGHTS

Overall attrition is around 16%, meaning nearly one in every six employees leaves the company.

Sales Representatives and Laboratory Technicians are among the highest-risk roles.

Attrition is highest for employees under 25 and within their first 0–3 years at the company.

Senior roles (Directors, Managers) have relatively low attrition.

Employees earning less than ₹5,000 per month have the highest attrition, while higher salary groups show significantly better retention.

Employees working overtime are nearly three times more likely to leave than those who do not work overtime.

Frequent business travel and longer commuting distances are associated with higher employee turnover.

Poor work-life balance and low relationship satisfaction are linked to higher attrition.

Factors with comparatively high attrition rate are:

1. Department
2. Salary
3. Age
4. Overtime

MULTI FACTOR ANALYSIS (based on top 4 factors)

```
132 • SELECT
133     Department,
134     Salary_Slab,
135     OverTime,
136     Age_band,
137     COUNT(*) AS Total_Employees,
138     SUM(Attrition = 'Yes') AS Attrition_Count,
139     ROUND(SUM(Attrition = 'Yes') * 100.0 / COUNT(*), 1) AS Attrition_Rate
140 FROM employee_profile
141 GROUP BY Department, Salary_Slab, OverTime, Age_band
142 HAVING COUNT(*) >= 10
143 ORDER BY Attrition_Rate DESC;
```

Result Grid							
Filter Rows:		Export:		Wrap Cell Content:			
	Department	Salary_Slab	OverTime	Age_Band	Total_Employees	Attrition_Count	Attrition_Rate
▶	Sales	<5000	Yes	<25	11	8	72.7
	Research & Development	<5000	Yes	<25	20	13	65.0
	Research & Development	<5000	Yes	25-34	67	33	49.3
	Sales	5000-9999	Yes	25-34	23	9	39.1
	Sales	<5000	No	<25	16	6	37.5
	Sales	<5000	Yes	25-34	22	7	31.8
	Research & Development	<5000	No	55+	10	3	30.0
	Sales	5000-9999	Yes	35-44	28	8	28.6
	Human Resources	<5000	No	25-34	14	4	28.6
	Research & Development	<5000	Yes	35-44	52	14	26.9
	Sales	5000-9999	No	45-54	18	4	22.2
	Sales	<5000	No	25-34	60	13	21.7
	Sales	10000-14999	No	45-54	10	2	20.0
	Sales	10000-14999	No	35-44	10	2	20.0
	Research & Development	<5000	No	<25	47	9	19.1
	Research & Development	5000-9999	Yes	25-34	18	3	16.7
	Research & Development	<5000	No	25-34	185	24	13.0
	Sales	5000-9999	No	25-34	74	9	12.2
	Research & Development	10000-14999	No	45-54	22	2	9.1
	Sales	<5000	No	45-54	12	1	8.3

Result 119 ×

INSIGHTS

- Attrition is driven by the combined effect of multiple factors rather than any single characteristic.
- Young employees (<25) in the Sales department, earning less than ₹5,000 and working overtime, represent the highest-risk employee segment, with a 72.7% attrition rate.
- The analysis shows that factors such as low salary and overtime amplify attrition risk when they occur together.

RECOMMENDATIONS

Based on the analysis above:

1 Focus on Sales & High-Risk Roles

- Prioritize **Sales Representatives, Sales Executives, Lab Technicians,** and **HR staff** for retention strategies.
- Conduct focused **pulse surveys** or small focus groups in these roles to understand:
 - Workload and stress levels
 - Pay/commission competitiveness
 - Managerial support and team culture
- Review **compensation structures, targets, and career paths** in Sales.

2 Improve Early Tenure Experience (0–3 Years)

- Introduce or strengthen:
 - Structured onboarding programs (first 90 days).
 - Buddy/mentorship programs for new hires.
 - Clear development plans within the first 6–12 months.
- Implement “**stay interviews**” at key milestones (e.g., 6 months, 1 year, 2 years) for early-career employees to surface issues before they resign.

3 Retain Younger Employees (<25 and 25–34)

- For younger employees, focus on:
 - Visible career paths and promotion criteria.
 - Training and skill development opportunities.
 - Rotational programs or internal mobility to keep work interesting.
- Consider whether expectations set during recruitment match the actual job.

4 Monitor and Address Demographic Patterns

- Use the **Gender** and **MaritalStatus** slicers to check for any groups with systematically higher attrition.
- If any disparities are found, investigate:
 - Work–life balance
 - Flexibility policies
 - Manager behavior or unconscious bias
- Ensure retention initiatives are inclusive and sensitive to different employee needs.

5 Establish Ongoing Monitoring

- Treat this dashboard as a **live tool**, not a one-off analysis.
- Refresh the data monthly or quarterly and track:
 - Overall attrition rate
 - Attrition in Sales and other high-risk roles
 - Attrition in the <25 and 0–3 years segments
- Set specific targets, e.g., “Reduce Sales Representative attrition from ~40% to <25% within 12 months”.